

HW 5

WH Location problem

Homework

Example (Con.) WH Location problem

Decision variable $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ location of the warehouse

Euclidean distance (ℓ_2 norm)

$$\text{Minimize} \quad \sum_{j=1}^4 r_j \|\mathbf{u}_j - \mathbf{v}\|_2 = \sum_{j=1}^4 r_j \sqrt{(x_j - x)^2 + (y_j - y)^2}$$

Rectilinear distance (ℓ_1 norm)

$$\min_{x,y} f(x,y) = \sum_{j=1}^4 r_j \|\mathbf{u}_j - \mathbf{v}\|_1 = \sum_{j=1}^4 r_j (|x_j - x| + |y_j - y|)$$

To linearize, we introduce auxiliary variables s_j and t_j

$$\begin{aligned} \min_{x,y,s_j,t_j} \quad & \sum_j r_j (s_j + t_j) \\ & s_j \geq x_j - x, \quad s_j \geq -(x_j - x) \quad \forall j \\ & t_j \geq y_j - y, \quad t_j \geq -(y_j - y) \quad \forall j \end{aligned}$$

We want to find the warehouse location (x,y) that minimizes the total yearly distance. Here, we use the *rectilinear* distance instead of the Euclidean distance.

$$\min_{x,y} 200(|5 - x| + |10 - y|) + \dots + 300(|12 - x| + |0 - y|)$$

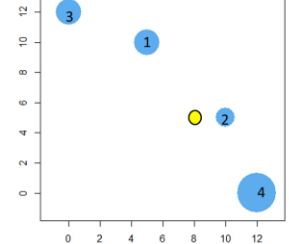
1. Linearize the problem.
2. Solve for the optimal warehouse location.

Revisit Slide05: Euclidean distance

Customer	Lat	Long	Distance	# of shipments per year
1	5	10	6.58	200
2	10	5	6.69	150
3	0	12	11.63	200
4	12	0	5.70	300

WH location $x_w = 9.11, y_w = 5.89$

Total yearly distance from WH to customers 5,456.54



	A	B	C	D	E	F	G	H	I	J	K
1	Customer (j)	Lat (Xj)	Long (Yj)	# of shipments per year (Rj)	xj-x	(xj-x)	yj-y	(yj-y)	Sj	Tj	Rj(Sj+Tj)
2	1	5	10	200	-5	5	5	-5	5	5	2000
3	2	10	5	150	0	0	0	0	0	0	0
4	3	0	12	200	-10	10	7	-7	10	7	3400
5	4	12	0	300	2	-2	-5	5	2	5	2100
6	WH location	10.0	5.0								
7	Total yearly distance from WH to customers				7500						
8											

1. Linearize the problem.

To linearize, we introduce auxiliary variables s_j and t_j

$$\begin{aligned} \min_{x,y,s_j,t_j} \quad & \sum_j r_j (s_j + t_j) \\ & s_j \geq x_j - x, \quad s_j \geq -(x_j - x) \quad \forall j \\ & t_j \geq y_j - y, \quad t_j \geq -(y_j - y) \quad \forall j \end{aligned}$$

2. Solve for the optimal warehouse location.

จากการแก้สมการจะได้ว่า เมื่อคำนวณระยะทางด้วย rectilinear distance เราควรตั้ง warehouse ที่พิกัด lat-long คือ $x=10$ และ $y=5$ เพื่อให้ได้ระยะเดินทางรวมที่น้อยที่สุดซึ่งจะเท่ากับ 7,500 miles